

Manual J Report



Prepared for Praestholm Residence (Rasmus)

643 Mt Pleasant Ave, West Orange, NJ 07052, USA

Created: Aug 12, 2026

Performed by: njenergyauditor@icloud.com

Calculations approved by ACCA to meet all Manual J 8v2.5 requirements

Heat: **40,901** Btuh

Cool: **25,108** Btuh

Heating Airflow: **1,246** CFM

Cooling Airflow: **1,031** CFM

Sensible Cooling: **22,571** Btuh

Latent Cooling: **2,537** Btuh

Sensible Ratio (SHR): **0.90**

Design Conditions

Weather Station: **ESSEX
COUNTY AP**

#724094, ASHRAE 2021

Latitude: 40.876

Elevation: 173 ft

Elevation Factor: 0.995

⚡ Indoor Heating 70°F

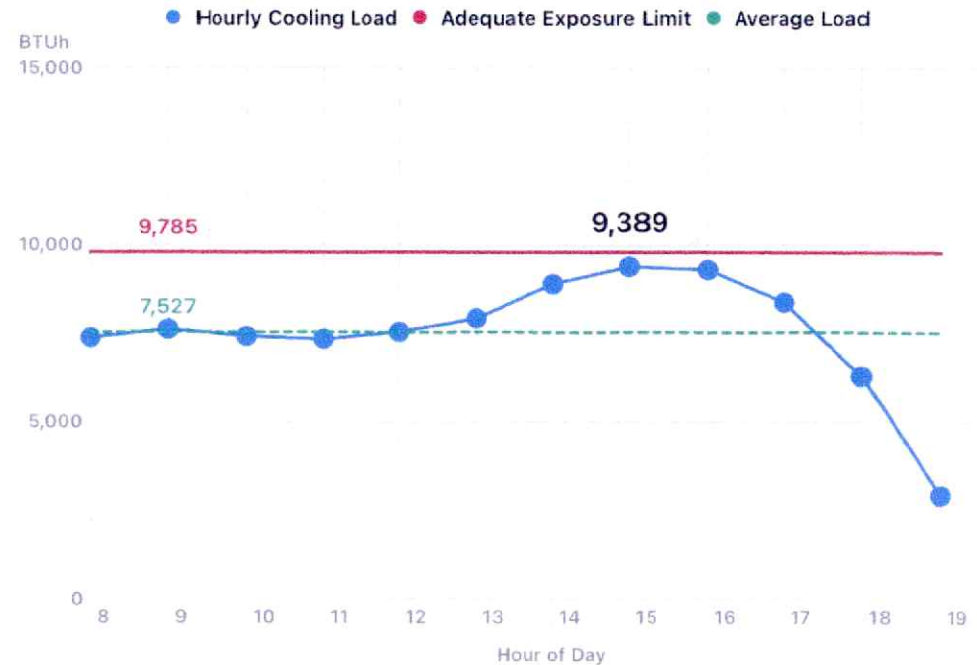
⚡ Indoor Cooling 75°F

⚡ Outdoor Heating 13°F

⚡ Outdoor Cooling 89°F

⚡ Design Grains 27.805

⚡ Daily Range: **Medium**



This application has adequate exposure diversity. The glass areas face a diverse set of directions and/or have sufficient shading (internal or external) to avoid solar gain peaks throughout the day.

Progression 'C'
Scope + Additional
Blower Door w/ 2K reduction

Building Component Summary

Exposed Wall Area: 2,063 sq ft | Conditioned Above Grade Volume: 12,664 cu ft

Component	Details (sq ft)	Heating	Sensible Cooling	Latent Cooling
Floors	978	1,172	97	0
Above Grade Walls*	1,543	9,492	3,958	0
Below Grade Walls*	679	8,804	1,583	0
Ceilings*	1,547	3,946	3,472	0
Windows	152	5,267	5,233	0
Skylights	0	0	0	0
Doors (glass)	43	1,301	1,219	0
Doors (opaque)	39	932	428	0
Infiltration	Blower Door (CFM50): 1751 CFM50 ACH4 Cooling: 0.00 ACH4 Heating: 0.00	9,987	1,323	1,537
Internal	Occupants: 5 Plants: 0 Appliance Scenario: medium	0	3,550	1,000
Envelope Loads Sub Total		40,901	20,864	2,537
Ventilation	None	0	0	0
Ducts	—	0	0	0
Blower Motor Heat	—	0	1,707	0
AED Excursion	This application has adequate exposure diversity	0	0	0
Grand Total (Btuh)		40,901	22,571	2,537

*Net surface area accounting for doors, windows and skylights

Construction Details (Building Level)

Component	Description Construction ID	Qty	Net Area (sq ft)	Detail	HTM
Floors	Passive fir over tight crawl space/bsmnt(R11Wall) - No floor insul, any finish 19C-0sp	1	264.5	U: 0.368	1.38 0.37
	Bsmnt fir (> 2' below grade) - 32' short slab edge, no insul, any finish 21A-32	3	713.9	U: 0.020	1.13 0.00
Above Grade Walls	2x4 Wood Framed wall, Stucco/Siding - R-0 cavity/R-0 board 12A-0s w	3	247.9	U: 0.240	13.58 6.36
	2x4 Wood Framed wall, Stucco/Siding - R-13 cavity /R-0 board 12C-0s w	13	762.1	U: 0.091	5.15 1.96
	Garage Partition 2x4 Wood Framed wall, Stucco/Siding - R-13 cavity /R-0 board 12C-0s w	4	222.0	U: 0.091	5.15 2.89
	2x4 Wood Framed wall, Stucco/Siding - R-13 cavity /R-6 board 12C-6s w	16	310.7	U: 0.060	3.40 0.80
Below Grade Walls	Avg depth: 3 ft Bsmnt 8" Stone/Brick Wall - R-0 Board, No Framing, No Finish 15B0-0-x	2	186.8	U: 0.293	16.58 1.68
	Avg depth: 4 ft Bsmnt 8" Stone/Brick Wall - R-0 Board, No Framing, No Finish 15B0-0-x	2	215.4	U: 0.297	16.82 2.73
	Avg depth: 3 ft Bsmnt 8" Stone/Brick Wall - R-0 Board, Wd Frm R-11, Finished 15B11-0w-x	1	132.9	U: 0.088	4.98 0.68
	Avg depth: 3 ft Garage Partition Bsmnt 8" Stone/Brick Wall - R-0 Board, Wd Frm R-11, Finished 15B11-0w-x	1	143.7	U: 0.086	4.87 1.69
Ceilings	Ceiling Under FHAvented w/o RB attic - R19 insul 16B-19	7	1,025.3	U: 0.049	2.77 2.44
	Ceiling Under FHAvented w/o RB attic - R21 insul 16B-21	1	271.6	U: 0.044	2.49 2.19
	Ceiling Under FHAvented w/o RB attic - R30 insul 16B-30	1	209.9	U: 0.032	1.81 1.59
	Ceiling Under FHAvented w/o RB attic - R56 insul 16B-50	1	40.4	U: 0.020	1.13 1.00
Windows	Northeast Clear Single pane operable window/door 1A-c Overhang: 0.00 ft Top to overhang: 1.00 ft Shade: Blind 45 deg slats, Medium Insect screen: None Sun screen: None Foreground reflectance: Crushed rock	1	3.6	U: 0.900 SHGC: 0.64	50.94 35.72
	Northwest Clear Single pane window w/storm 1C-c Overhang: 0.00 ft Top to overhang: 1.00 ft Shade: Blind 45 deg slats, Medium Insect screen: None Sun screen: None Foreground reflectance: Crushed rock	1	9.1	U: 0.570 SHGC: 0.56	32.26 28.04
	Northeast LowE Double pane operable window/door 2A Overhang: 0.00 ft Top to overhang: 1.00 ft Shade: Blind 45 deg slats, Medium Insect screen: None Sun screen: None Foreground reflectance: Crushed rock	3	19.7	U: 0.550 SHGC: 0.56	31.13 28.40
	Northwest LowE Double pane operable window/door 2A Overhang: 0.00 ft Top to overhang: 1.00 ft Shade: Blind 45 deg slats, Medium Insect screen: None Sun screen: None Foreground reflectance: Crushed rock	1	9.8	U: 0.550 SHGC: 0.56	31.13 28.40
	South LowE Double pane operable window/door 2A Overhang: 0.00 ft Top to overhang: 1.00 ft Shade: Blind 45 deg slats, Medium Insect screen: None Sun screen: None Foreground reflectance: Crushed rock	2	20.9	U: 0.550 SHGC: 0.56	31.13 20.72
	Southeast LowE Double pane operable window/door 2A Overhang: 0.00 ft Top to overhang: 1.00 ft Shade: Blind 45 deg slats, Medium Insect screen: None Sun screen: None Foreground reflectance: Crushed rock	3	26.6	U: 0.550 SHGC: 0.56	31.13 34.79
	Southwest LowE Double pane operable window/door 2A Overhang: 0.00 ft Top to overhang: 1.00 ft Shade: Blind 45 deg slats, Medium Insect screen: None Sun screen: None Foreground reflectance: Crushed rock	2	10.7	U: 0.550 SHGC: 0.56	31.13 34.79
	West LowE Double pane operable window/door 2A Overhang: 0.00 ft Top to overhang: 1.00 ft Shade: Blind 45 deg slats, Medium Insect screen: None Sun screen: None Foreground reflectance: Crushed rock	2	18.5	U: 0.550 SHGC: 0.56	31.13 40.55
	Southeast LowE Double pane operable window/door 3A Overhang: 0.00 ft Top to overhang: 1.00 ft Shade: Blind 45 deg slats, Medium Insect screen: None Sun screen: None Foreground reflectance: Crushed rock	1	6.8	U: 0.530 SHGC: 0.56	82.49 66.99
	Southwest LowE Double pane operable window/door 3A Overhang: 0.00 ft Top to overhang: 1.00 ft Shade: Blind 45 deg slats, Medium Insect screen: None Sun screen: None Foreground reflectance: Crushed rock	1	26.8	U: 0.530 SHGC: 0.56	34.50 39.67
Skylights	-	0	0	-	0 0
Doors (glass)	Northeast LowE Double pane operable window/door 3A Overhang: 0.00 ft Top to overhang: 1.00 ft Shade: Blind 45 deg slats, Medium Insect screen: None Sun screen: None Foreground reflectance: Crushed rock	1	43.4	U: 0.530 SHGC: 0.56	30.00 28.10
Doors (opaque)	Wood - Solid core w/metal storm 11F	1	21.5	U: 0.280	15.85 7.28
	Metal - Fiberglass core 11J	1	17.4	U: 0.600	33.96 15.60

● Heating HTM ● Cooling HTM

Below Grade

Basement Storage Side

Heat: 2,737 Btuh
Cool: 797 Btuh

Latent Cooling	99
Sensible Cooling	698
Sensible Ratio (SHR)	0.88
CFM Heating	83
CFM Cooling	32

Basement Utility Side

Heat: 9,564 Btuh
Cool: 1,896 Btuh

Latent Cooling	291
Sensible Cooling	1,605
Sensible Ratio (SHR)	0.85
CFM Heating	291
CFM Cooling	73

Basement Bathroom

Heat: 575 Btuh
Cool: 183 Btuh

Latent Cooling	15
Sensible Cooling	168
Sensible Ratio (SHR)	0.92
CFM Heating	18
CFM Cooling	8

1st Story

Exterior Front Picture

Heat: 4 Btuh
Cool: 3 Btuh

Latent Cooling	0
Sensible Cooling	3
Sensible Ratio (SHR)	1.00
CFM Heating	0
CFM Cooling	0

1F Bathroom

Heat: 590 Btuh
Cool: 343 Btuh

Latent Cooling	27
Sensible Cooling	316
Sensible Ratio (SHR)	0.92
CFM Heating	18
CFM Cooling	14

Front Corner Bedroom

Heat: 6,652 Btuh
Cool: 3,591 Btuh

Latent Cooling	286
Sensible Cooling	3,305
Sensible Ratio (SHR)	0.92
CFM Heating	203
CFM Cooling	151

Living Room & Foyer / Hallway

Heat: 3,763 Btuh
Cool: 2,592 Btuh

Latent Cooling	104
Sensible Cooling	2,489
Sensible Ratio (SHR)	0.96
CFM Heating	115
CFM Cooling	114

Rear Corner Bedroom

Heat: 2,717 Btuh
Cool: 1,663 Btuh

Latent Cooling	127
Sensible Cooling	1,537
Sensible Ratio (SHR)	0.92
CFM Heating	83
CFM Cooling	70

Kitchen & Dining

Heat: 7,137 Btuh
Cool: 4,169 Btuh

Latent Cooling	185
Sensible Cooling	3,985
Sensible Ratio (SHR)	0.96
CFM Heating	217
CFM Cooling	182

2nd Story

2/FA Cape Stairwell

Heat: 239 Btuh

Cool: 91 Btuh

Latent Cooling	17
Sensible Cooling	74
Sensible Ratio (SHR)	0.81
CFM Heating	7
CFM Cooling	3

Cape Bathroom

Heat: 422 Btuh

Cool: 214 Btuh

Latent Cooling	23
Sensible Cooling	191
Sensible Ratio (SHR)	0.89
CFM Heating	13
CFM Cooling	9

Cape Right Side Bedroom

Heat: 2,558 Btuh

Cool: 1,213 Btuh

Latent Cooling	156
Sensible Cooling	1,057
Sensible Ratio (SHR)	0.87
CFM Heating	78
CFM Cooling	48

Office, Etc

Heat: 3,943 Btuh

Cool: 2,094 Btuh

Latent Cooling	208
Sensible Cooling	1,886
Sensible Ratio (SHR)	0.90
CFM Heating	120
CFM Cooling	86

General

Stories 2

Bedrooms 4

Estimated Living Area 1,604 sq ft

Year Built 1946

Attached Garage: **Open**
minutes per day

Natural gas? ☒

Existing heat pumps? ☐

Conditioned Basement? ☒

Room over garage? ☐

HVAC System Type: **Duc...**

Building Type: **Single Family**

Internal

Occupants 5

Scenario: **2400 btuh - Standard Kitchen, utility and lighting**

Heat: **0** Btuh
Cool: **4,550** Btuh

Latent Cooling 1,000
Sensible Cooling 3,550
Sensible Ratio (SHR) 0.78

Plants 12" or shorter 0

Plants 12" - 24" 0

Plants 24" or taller 0

Infiltration

Method: **Blower Door (C...**

Leakage 1,751 CFM50

4 - Substantial number of obstructions shield most of the perimeter: buildings, hedges, solid fences — typical suburban shielding

Shielding Class: **4**

Heat: **9,987** Btuh
Cool: **2,860** Btuh

Latent Cooling 1,537
Sensible Cooling 1,323
Sensible Ratio (SHR) 0.46

Ventilation

Type: **None**

Heat: **0** Btuh
Cool: **0** Btuh

Latent Cooling 0
Sensible Cooling 0
Sensible Ratio (SHR) 0.00

Cubic Feet Per Minute of air flow @ 50 pa (CFM50) is the volumetric flowrate of air as measured by a blower door test at 50 pascals of pressure differential.

At Audit BA was 3751 w/ACH
Scanned Living Area: 2,261 sq ft
Estimated Living Area: 1,604 sq ft